Course syllabus for

Academic year 2024/2025 >

TIF295 - Experimental methods in modern physics

Experimentella metoder inom modern fysik

Course syllabus adopted 2023-02-07 by Head of Programme (or corresponding)
Owner: MPPHS
3,0 Credits
Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail
Education cycle: Second-cycle
Main field of study: Engineering Physics
Department: 16 - PHYSICS

Course round 1

Teaching language: English
Application code: 85116
Open for exchange students: Yes
Block schedule: A
Maximum participants: 60 (at least 10% of the places are reserved for exchange students)

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course	No Sp	Examination dates
0119 Project 3,0 c Grading: TH	3,0 c						

In programs

MPCAS COMPLEX ADAPTIVE SYSTEMS, MSC PROGR, Year 1 (compulsory elective)
MPCAS COMPLEX ADAPTIVE SYSTEMS, MSC PROGR, Year 2 (elective)
MPPHS PHYSICS, MSC PROGR, Year 1 (compulsory)

Examiner:

Lena Falk

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Eligibility

General entry requirements for Master's level (second cycle)
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

English 6 (or by other approved means with the equivalent proficiency level)
Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Course specific prerequisites

Basic courses in physics

Aim

The purpose of this course is to introduce the students to a variety of experimental concepts used in modern physics research, and to give them an opportunity to work on a well defined experimental project of their own choice. The projects span over a wide range of disciplines, from biophysics and materials physics to nanooptics and surface physics, and will be carried out in research groups working in state-of-the-art laboratories at the Department of Physics. The students will work in groups of two or three, and the projects will be presented both in writing and as an oral presentation in a seminar. As a general introduction to the topic, the course will start with several lectures on the physics behind the projects that the students may choose from.

Learning outcomes *(after completion of the course the student should be able to)*

After this course the students should have a general knowledge of experimental methods in modern physics. The students should be able to write a scientific report based on their own experimental work, and to give an oral presentation describing the used method and the obtained results. After this course the students should also have an increased knowledge and awareness of gender equality, equal treatment and diversity, which will give them the ability to challenge and change unequal structures and cultures.

Content

The course starts with a number of lectures on the physics behind the different experimental projects that the students may choose from. The projects are then carried out in groups of two or three students.There will also be a lecture on entrepreneurship, and this topic is then addressed in the laboratory projects. Gender equality, equal treatment and diversity will also be discussed in this course. There will be one lecture on these topics in the beginning of the course, and the students are encouraged to reflect on these questions during the work on their project. Finally, the students write a report and give an oral presentation.

Organisation

Lectures, an experimental project, report writing, oral presentation.

Literature

Reading material in the form of lecture notes and laboratory instructions, including references to the literature, will be available on the course home page.

Examination including compulsory elements

The students carry out an experimental project, and hand in a written report on their work. The report is presented orally in a seminar. A contribution report, handed in together with the project report, should describe the different group members contributions to the project and the writing of the project report, and discuss the work crried out in the group from a gender equality, equal treatment and diversity perspective.

The course examiner may assess individual students in other ways than what is stated above if there are special reasons for doing so, for example if a student has a decision from Chalmers on educational support due to disability.